ChatGPT

Shared Containerized Waste Collection: Lessons from Barcelona and Beyond

Introduction: What is a "Shared Container" System?

In a **shared containerized waste collection system**, residents bring their garbage and recyclables to **communal bins** placed in public areas (on streets or underground), instead of each household having individual curbside bins. These large containers (often sorted by waste type) are periodically emptied by the city. This approach is common in many dense cities in Europe and Asia, aiming to streamline collection, keep streets cleaner, and encourage recycling. Below, we review Barcelona's model – often cited as a reference – and compare it with similar systems in other cities (San Francisco, Vienna, Seoul, Amsterdam). We evaluate design features, operations, costs, environmental outcomes, social equity issues, and challenges, distilling insights relevant to a potential rollout in Washington, D.C.

Barcelona's Communal Container System

Barcelona's street-side waste containers for separate collection of glass (green), paper (blue), plastic/metal (yellow), organics (brown), and other waste (gray) [1]. Residents walk their sorted trash to these color-coded bins, which are strategically placed on sidewalks near buildings.

Design and Infrastructure in Barcelona

Barcelona has long utilized a **communal container ("Área de Acera") model**, where sets of large, color-coded bins are stationed on public streets for different waste streams [2] [1]. Typically, five containers are

grouped together: **green for glass, blue for paper/cardboard, yellow for plastic & metal packaging, brown for organics, and grey for general refuse** [1]. This standardized color system is used across Catalonia and Spain, making it intuitive for residents and visitors alike. Bins are placed to serve a cluster of buildings, often on each block or neighborhood corner, so that most residents have a drop-off point within a short walk.

To improve functionality and street aesthetics, Barcelona has experimented with **advanced container designs**. New bins rolled out in 2022 are **lower in height** (to preserve pedestrians' sightlines and safety) and come in **various sizes** to fit different street layouts [3]. The container openings are calibrated for the intended waste type (for example, smaller openings for glass or cans) to prevent misuse [3]. In some areas (particularly historic or high-density zones), Barcelona and other Spanish cities have also installed **underground containers**: bins are buried below the street with only a chute or pillar above-ground, keeping sidewalks clearer and trash out of sight [4]. A few districts even use **pneumatic collection** (vacuum tubes that whisk trash from street inlets to a central facility), which reduces noise and surface traffic [5], although this system is limited to select locations due to high infrastructure cost [6].

Notably, Barcelona is moving toward **"smart" containers** with user identification. In one pilot, *organics* bins were outfitted with RFID/chip-enabled locks that only open when a resident taps a personal key fob [7] [8]. This allows the city to track participation and volume by household. The smart brown bins were introduced after a 2021–2022 trial of door-to-door collection met resistance in Sant Andreu; the tech-enabled shared bins let residents dispose of organics at any time (instead of during limited pickup hours), while still enabling individual accountability [7] [9]. Each smart bin cost about **€1,400**, roughly €400 more than a standard container [10] due to the added electronics. Barcelona's long-term plan is to gradually replace traditional bins with smart containers citywide, paving the way for future **pay-as-you-throw** schemes and targeted incentives [11] [12].

Operations and Collection Logistics

Under Barcelona's current system, city sanitation crews service the shared containers on a regular schedule. **Collection trucks** (many now electric or hybrid for reduced noise and emissions [13]) either lift and tip the bins or, in the case of underground units, use a **hydraulic crane** to hoist the buried container up and empty it. Collection is typically **daily for organics and refuse**, and several times per week for recyclables, though frequencies are adjusted based on neighborhood generation rates and season. The city staggers pick-up times (often at night for refuse/organics to minimize daytime traffic and odors [14]).

Because containers consolidate waste from many households, the **collection process is more automated** than door-to-door service. A single truck can mechanically empty a series of containers in far fewer stops than collecting individual bins or bags at every address. Barcelona's 2022 upgrade contract significantly increased the electric collection fleet (from 20% to 66% electric vehicles) and introduced onboard technology (sensors/GPS) to optimize routes and monitor bin fill levels in real time [13] [15]. Some containers are equipped with **ultrasonic fill sensors** that alert crews when nearing capacity [16]. These efficiencies can reduce staffing needs per ton collected – one technical study noted that underground/communal systems yield **staff cost savings** compared to traditional methods [17] [18]. However, those savings may be offset by the need for specialized trucks and maintenance of more complex infrastructure.

Barcelona's operational model is still evolving by district. In most areas, communal bins handle all household waste. In a few historic neighborhoods with very narrow streets, the city piloted **door-to-door**

pick-up (termed *"porta a porta"*) where residents set out small bins or bags on a schedule instead of using street containers [19]. This was done to improve recycling rates, but it required intensive staffing and strict adherence to schedules, and it faced community pushback (as discussed later). As a result, Barcelona is likely to **apply different collection models in different zones**, depending on urban form: dense urban blocks may keep shared containers, while low-rise or hard-to-access streets might use door-to-door or block collection points [20] [21]. The key for operations is flexibility – matching the collection method to neighborhood needs while maintaining overall efficiency.

Economic Efficiency: Costs vs. Traditional Curbside

The economic efficiency of shared container systems in Barcelona has been a subject of analysis and debate. On one hand, communal bins allow **more efficient routing and fewer collection stops**, reducing labor hours and truck fuel per amount of waste. In cities that have adopted underground or automated container systems, officials report significant **operational savings** – for example, a European study found an underground container network can be **36% less expensive than door-to-door collection** in operating costs [22]. Barcelona's own trials echoed this: personnel costs dropped as fewer workers were needed to cover the same area [17].

On the other hand, these savings are balanced by **high upfront investments**. Purchasing and installing robust containers (especially underground or "smart" models) and retrofitting trucks with cranes or lifting mechanisms is costly [23]. Barcelona's newest waste contract (2022–2030) is its largest ever – €2.3 billion over 8 years – largely to fund new equipment and system upgrades [24]. Maintenance costs for high-tech bins (locks, sensors, etc.) are also non-trivial [25].

Crucially, **economies of scale** determine whether the system truly saves money. A 2025 analysis concluded that communal container systems can achieve overall costs *comparable to or lower than* door-to-door collection **only if implemented citywide (or across a large area)** [26] [27]. If they are just layered on top of existing services or used in parallel with curbside pickup, the redundancy can eliminate any cost advantage [28]. Barcelona's experience reflects this: running a dual system (communal bins in some districts, door-to-door in others) meant maintaining two sets of operations, which is inefficient. Thus, from an economic standpoint, **full commitment and coverage** may be needed to reap savings.

User costs in Barcelona are currently a flat annual fee, but the introduction of smart containers opens the door for a fairer fee structure. In Sant Andreu's pilot, residents who logged over 40 organics disposals per year (using their personal bin fob) received a **30% rebate on their waste fee** [11] – effectively rewarding those who separate organics diligently. This hints at future **pay-as-you-throw (PAYT)** models where households pay according to the waste they generate or the frequency they use the bins. Such schemes can improve economic efficiency by internalizing the cost of waste disposal and providing direct incentives to reduce waste. However, implementing PAYT would require the entire city's infrastructure to support user identification (one reason Barcelona is phasing in smart bins gradually).

In summary, Barcelona's communal system involves higher capital and technology costs than traditional curbside collection, but it promises lower marginal operating costs once fully rolled out. Other cities have reported long-term savings and even reduced disposal costs (due to higher recycling) with containerization [22]. For Washington, D.C., the lesson is that up-front investment and unified deployment are necessary to achieve cost efficiency; a half-measure or mixed approach might end up costing more rather than less.

Environmental and Cleanliness Impacts

One of the driving goals for Barcelona's shift in waste collection methods is improving environmental outcomes – both in terms of **cleaner streets** and **higher recycling rates**. The communal container approach has shown mixed results on these fronts, with notable improvements in some areas and new challenges in others.

Street cleanliness: Switching from individual trash bags on the curb to sealed containers generally keeps public spaces cleaner. In cities where garbage was traditionally left out in bags (exposed to pests and weather), installing containers eliminated the routine spillage and **mess on pavements** that occurred on collection days [17]. Barcelona, which decades ago had curbside bag collection in parts of the old city, saw those streets become tidier once communal bins were introduced – waste is contained and not ripped open by animals. An analysis across several European cities likewise found that underground containers **improved cleanliness of sidewalks** compared to bag collection [17]. Keeping waste mostly enclosed (and even underground in some cases) also reduces odors and deters rodents/insects.

However, communal bins bring the issue of **illegal dumping or overflow**. If a container is full or locked, some residents will leave garbage bags **on the ground next to the bin**, creating an eyesore and public health concern. Barcelona has faced this, especially in high-density areas or during holidays when waste volume spikes. The city responded by increasing collection frequency and adding capacity (e.g. *25% more recycling containers* in the 2022 upgrade [29]) to prevent overflow. They also deployed surveillance and fines in chronic dumping spots. Other cities have encountered similar issues: a study noted **dumping around collection points** as a persistent problem with shared bins, even as overall street cleanliness improved (no more piles of bags every night) [30]. This suggests that **management of container capacity and user compliance** is key – technology like fill sensors and prompt emptying, as well as public education, can mitigate container overflow.

Recycling and waste diversion: Barcelona's communal bins have historically yielded **moderate recycling rates**, but not as high as desired. As of the early 2020s, the city's overall selective collection rate was around **38%** [31] – meaning the majority of waste was still ending up in the grey "rest" container. The convenience of 24/7 drop-off helps encourage recycling (since bins for each stream are readily available), but it can also lead to contamination if people are not careful about sorting. For example, having open access to an organics or recycling bin might tempt some to toss improper items (especially if the general bin is far or if PAYT rules make disposing of trash more costly [32]). Indeed, when a pay-per-use trial was implemented elsewhere, **organics bins saw high contamination** from people sneaking trash into them to avoid fees [32]. Barcelona has been mindful of this and thus introduced the *user-ID locks on organic bins* to discourage abuse [9] – only authorized residents can open them, and improper use can be traced.

In pursuit of EU recycling targets, Barcelona experimented with the door-to-door system in some neighborhoods, which dramatically improved diversion – one pilot area's recycling rate jumped from about **40% to 80%** after door-to-door separation was enforced [33] [31]. This underscores a key point: **communal drop-off alone might not maximize recycling without additional measures**. Barcelona's approach now is to marry the convenience of shared containers with **policy nudges**: making certain bins (like organics) accessible only to those who sort properly, and potentially charging or rewarding households based on their sorting performance [11]. The city also relocated bins so that recycling options (blue, yellow, brown) are always co-located and just as convenient as the grey bin [34], to remove any excuse for dumping everything in general waste.

Secondary environmental benefits: A less visible but important impact is on emissions and noise. By optimizing routes and reducing the number of collection stops (thanks to aggregated containers), garbage trucks spend less time idling and driving; this lowers fuel consumption and thus CO_2 emissions. Barcelona's move to electrify two-thirds of its fleet amplifies this benefit, cutting air pollution from diesel trucks [13]. Noise pollution is also improved: modern sealed containers and electric lifter mechanisms are quieter than the clattering of collecting dozens of individual bins or bags. Residents have complained about nighttime collection noise in the past, so the city invested in **sound-dampening truck equipment** and schedules noisy operations (like glass collection) carefully [5] [13]. Pneumatic systems, though rare, virtually eliminate collection vehicle noise on the street by using underground vacuum tubes [5].

In summary, Barcelona's container system has **made streets cleaner overall** by containing waste and supporting frequent collection, but it requires diligent management to prevent new messes (like pile-ups around bins). Recycling outcomes have improved slowly; the city learned that **infrastructure alone isn't enough** – pairing the shared bins with smart tech, incentives, and community engagement is necessary to boost waste separation rates toward their goals. For D.C., this highlights that a containerized system can yield environmental gains (less litter, potentially more recycling) if well-maintained and coupled with education and enforcement to ensure proper usage.

Social and Equity Considerations

Implementing shared waste containers can have complex social impacts – some positive, some raising equity concerns. **Who benefits and who bears any burdens** in Barcelona's system has been an ongoing discussion, especially as the city tries new collection models in different neighborhoods.

One benefit of communal bins is **convenience and flexibility**. Residents are not tied to a collection schedule; they can take out trash at any time that suits them (Barcelona encourages doing so in the evening to minimize odors [14], but it's not strictly enforced). Many people appreciate not having to remember garbage day or store smelly waste for a week – an advantage noted in other cities with 24/7 drop-off systems [35]. In Seoul, for instance, residents valued that RFID food waste bins let them dispose of scraps daily and avoid bad smells at home [36] [37]. This on-demand aspect tends to benefit those in apartments or dense housing who lack space to keep multiple bins. It can also foster a **sense of communal responsibility** – seeing the neighborhood bins regularly can reinforce awareness of recycling in the public consciousness.

On the other hand, not everyone finds the system equitable. In Barcelona's door-to-door pilot, **older adults and people with limited mobility** found it challenging to comply with strict schedules and to carry bins outside within short time windows. The switch back to nearby shared containers in Sant Andreu was partly to accommodate those who struggled with the previous regime [7]. But even with communal bins, there's the issue of **distance and accessibility**: if a person lives on an upper floor with no elevator, or has a disability, taking trash down the block to a container could be difficult. A well-designed system will ensure **containers are frequent and conveniently located** (the 2025 study suggests having enough containers such that users are within a "reasonable perimeter" distance) [18]. In practice, Barcelona aims for no more than 100-150 meters from any residence to the nearest set of bins. Nonetheless, ensuring equal ease of use for the elderly or disabled might require accommodations, such as assisted services or building-level arrangements for those who can't easily use the public bins.

Aesthetic and social acceptance is another consideration. Large garbage containers on streets can be seen as unsightly or as detracting from neighborhood charm. Some Barcelona residents have complained

that bins (especially older models) are bulky "street clutter," occupy parking space, or block sidewalks in narrow areas [38]. The city responded by redesigning bins to be lower-profile and by tailoring container size to each street's context [3]. In very historic districts, the push has been to remove bins entirely from small medieval streets – either by using *underground bins* or by implementing door-to-door in those enclaves to preserve the streetscape. Community engagement is crucial: Barcelona learned this when the abrupt introduction of door-to-door in Sant Andreu sparked fierce opposition and even became a political flashpoint [19]. Officials had to backtrack and involve residents in planning a modified approach [39]. The lesson is that any major change to how people dispose of waste touches daily routines and public space, so **early consultation and responsiveness to concerns** (safety, smell, visibility) are needed to gain social license.

Equity in cost and enforcement is also important. A potential downside of communal drop-off is the feeling that "free riders" can occur – for example, if some neighbors don't sort their trash and contaminate recycling bins, everyone's efforts are undermined. Or if businesses use public residential bins to avoid commercial fees, locals bear the burden of overflow. Barcelona is tackling this by tracking usage: the smart container system essentially individualizes responsibility (you use your key to open the bin, your contributions are logged). This not only enables incentive rewards as mentioned (fee discounts for avid recyclers) [11], but could also identify violators if, say, someone routinely misuses bins. **Privacy and acceptance** of being "monitored" for trash disposal is an equity consideration too – some initially balked at having to identify themselves just to throw away garbage [40]. Clear communication that the goal is to improve service and fairness (and not to overly police residents) helped ease concerns.

Interestingly, communal systems can also highlight equity issues between neighborhoods. In wealthier or newer districts, apartment complexes might have private waste rooms or even their own compactors, meaning residents hardly see the garbage. In older or lower-income areas, the communal containers are very much in public view and can sometimes attract pests or litter if not serviced well. Ensuring **all areas get equal service frequency and container maintenance** is vital so that one community isn't bearing disproportionate negatives. Barcelona's new contract explicitly increased cleaning around containers and is hiring more sanitation workers (an **"incremento de operarios"**) to keep container areas tidy [41] – this benefits especially those dense neighborhoods where a spilled mess at a bin could affect many people.

Finally, one equity aspect is **who pays and who benefits financially**. If the city moves to PAYT, residents generating less waste or diligently recycling (often higher-income, more educated households) can reduce their fees, while larger households or those with less ability/time to sort might pay more – which could be seen as regressive if not designed carefully. On the flip side, the current flat fee system meant that even neighborhoods with poor recycling performance paid the same; introducing incentives or penalties is intended to be fairer (polluters pay principle). The key is to accompany it with support (like multilingual education, free recycling bags/bins, etc.) so that all groups can participate in waste reduction equally.

In summary, the shared container model in Barcelona offers **greater convenience and potential cost fairness**, but it must be implemented with sensitivity to accessibility and neighborhood concerns. Community buy-in improves when the system is adapted to local conditions (e.g. smaller bins in tight streets, underground options in historic areas) and when residents see direct benefits (cleaner streets, possible tax discounts). The experience suggests D.C. would need to actively manage these social factors – from placement of containers (not blocking sidewalks or sightlines) to programs that assist those who might struggle with the new system – to ensure equitable outcomes.

Reported Challenges and Drawbacks

No waste collection system is without its problems, and Barcelona's journey has highlighted several challenges that come with communal containers. Being aware of these issues is crucial for any city (like Washington, D.C.) considering a similar approach, so that mitigation strategies can be planned in advance.

1. Misuse and contamination: A common challenge is that some users do not follow the rules – they might dump inappropriate materials in or around the containers. Barcelona has seen instances of **businesses illegally dumping commercial waste** into neighborhood bins, which leads to overflow, and of residents disposing of trash in recycling or organics containers (especially if those are not locked). Such misuse reduces the efficiency of recycling (contaminated recyclables may end up landfilled) and creates extra clean-up work. Other cities echo this: In the Netherlands, communal street bins sometimes suffer from people forcing in oversize items or leaving bags at the base if the opening is too small or if the bin is locked until a card is used [42]. Clear labeling, public education, and surveillance can help, but it remains a behavioral challenge.

2. Littering and illegal dumping around bins: **As noted earlier, when containers fill up or someone can't be bothered to sort properly, they might abandon garbage next to the bin. This not only litters the area but can attract vermin. Amsterdam and other Dutch cities report a phenomenon of** "side waste" **– bags or junk left near underground containers – which city crews then have to collect separately. In some cases,** scavengers **exacerbate the issue: Amsterdam's city survey in 2025 found that people hunting deposit-refund bottles were prying open bins and dumping out contents, causing damage and street litter** [43] [44]**. The scale was significant enough that major Dutch cities face** millions in costs** from repairing and cleaning up bins due to this issue [44]. To combat it, measures such as sturdier, lockable container designs and adding separate receptacles specifically for deposit bottles (so people don't need to raid trash bins) were introduced [45] [46]. The takeaway is that cities must anticipate some degree of container tampering or dumping and have enforcement or design solutions ready (e.g. CCTV monitoring of notorious dump sites, rapid response teams to clear any side waste before it grows).

3. Noise and aesthetics: While containerization is generally cleaner, the **noise** of collection can be a nuisance. The clanging of metal bins or the beeping of trucks at night has led to complaints. Barcelona has partly addressed this by shifting to electric vehicles and *sound-insulated equipment* [13]. Still, emptying a large container (especially glass) can be loud. Some residents also find the presence of containers in front of their homes undesirable visually or worry about odor if a bin is right under their window. The city's approach has been to site bins considerately (not directly in front of cafe terraces or ground-floor windows if possible) and to wash containers frequently. **Regular cleaning** of the bin and its surrounding area is essential to prevent smells and pest infestations – this adds to operational tasks, but cities like Barcelona include it in the contract (e.g. power-washing containers on a set schedule). Aesthetically, a number of cities beautify communal bins with public art or camouflage. Amsterdam has decorated some above-ground bin units to blend into streetscapes [47], and Seoul's food waste bins are often kept in neat enclosures at apartment complexes to hide the mess. Despite these efforts, it's fair to say communal containers remain a visible piece of street furniture and not everyone will love that.

4. Technical and maintenance issues: Introducing smart containers or underground mechanisms means there is more that can break down. Barcelona's pilot saw a few **smart organics bins jam** or fail to unlock, which frustrated residents and led a few to simply leave their waste outside the bin. Likewise, underground container systems depend on hydraulic lifts and sensors – if a mechanism fails, that container might be out

of service and overflow until fixed. Cities need a robust maintenance program and budget for spare parts, software updates, etc. Smaller cities or those with tight budgets could struggle; in South Korea, after the central government stopped subsidizing RFID waste bins, some poorer municipalities had **trouble expanding or even maintaining** their smart bin programs [48]. This indicates a need for ensuring long-term funding and local technical expertise whenever high-tech solutions are implemented.

5. Parallel systems during transition: During the rollout phase, as mentioned, running two systems (old and new) concurrently can be problematic. Barcelona experienced this in Sant Andreu, where door-to-door and communal bins were both partially in use – confusing residents and doubling costs [28]. For Washington, if a pilot of communal bins is done, there should be a clear plan to scale up or integrate it, so neighborhoods aren't stuck in a costly hybrid mode for too long.

6. Enforcement and privacy: Lastly, enforcing proper use of shared bins can raise privacy or civil liberty concerns. Tracking waste disposal with personal IDs (as in Barcelona's smart bins or Seoul's RFID system) means **data is collected on individuals' habits**. While this can be used positively (to give rewards or pinpoint support where needed), some citizens may view it as intrusive. Ensuring data security and limiting the use of such data to just waste management purposes is important to maintain public trust. Additionally, actual "garbage police" (inspectors who might open bags to identify who dumped incorrectly) have been used in some places – San Francisco famously has strict enforcement for mandatory composting [49] – but that approach needs a careful balance so it doesn't come across as draconian or inequitable in enforcement.

In summary, the challenges of communal container systems include **human behavior issues (misuse, dumping, scavenging)**, **nuisance issues (noise, visuals)**, and **technical/administrative issues (maintenance and managing changeover)**. Barcelona's adaptive strategies – like container redesign, tech enhancements, increased cleaning, and incentive/fine structures – provide useful guidance on tackling these issues. Any city aiming to implement such a system should budget and plan not just for the containers and trucks, but also for the "soft" aspects: public outreach, enforcement personnel, and contingency plans for things like vandalism or system outages.

Comparisons to Other Cities' Systems

Barcelona's experience is insightful, but it's also important to see how other major cities have approached shared or containerized waste collection. Cities around the world have adopted variations on this model, each with their own successes and caveats. Below we examine a few: **Amsterdam**, **Vienna**, **Seoul**, and **San Francisco** – highlighting elements of their systems and any lessons learned that complement the Barcelona case.

Amsterdam (and the Netherlands): Underground Communal Bins

Amsterdam is often cited as a pioneer of modern communal container systems. In fact, **most neighborhoods in Amsterdam have no weekly "trash day" at all** – instead, residents take their garbage and recyclables to **sidewalk inlets that lead to large underground containers** [50]. Only a few outlying areas still use curbside wheelie bins. This system, widespread across the Netherlands, was introduced to improve urban cleanliness and efficiency.

Design: The typical setup is a small metal box or pillar on the sidewalk (for each waste stream: e.g. one for general waste, one for paper, one for glass, etc.), which is actually the top of a deep container buried below. The underground containers can hold **5+ times more waste** than an equivalent curb bin, and being sealed underground helps prevent pests and scavengers [51]. Many are equipped with **RFID access** for general trash: residents tap a key-card to unlock the bin and deposit their bag [52]. This allows municipalities to implement PAYT – some Dutch cities charge per trash bag dropped (tracked via the card) [42], creating a direct financial incentive to recycle more and throw out less. In Amsterdam, not all areas require a card; in some neighborhoods the general waste inlets are open access, while recycling bins (glass, paper, etc.) are usually open to all [52]. The containers themselves are heavy-duty and tamper-resistant, though as noted, **resourceful "deposit hunters" have occasionally managed to break into them** to retrieve bottles [43], prompting the city to add locks and even remove some problem bins [46].

Operations: Collection in Amsterdam is carried out by trucks with on-board cranes. When a container is due for emptying (either on a schedule or when a fill-level sensor alerts that it's near full), a crew arrives, parks at the site, and uses a hydraulic crane arm to lift the container out of the ground. The container is a large metal box that gets hoisted and then **emptied from the bottom into the truck's hopper**, after which it's lowered back into its silo [53]. This process is quite automated; often a single operator can do it, using a remote control to manage the crane. It's almost "futuristic" – one could describe the workers as *robot arm operators* rather than traditional garbage men [54]. The cycle per container is longer than a quick dump of a curb bin, but since each container holds so much and serves many households, the overall route efficiency is high (fewer stops overall). One challenge is noise – the crane operation and dumping of a whole container can make a loud bang. Amsterdam mitigates this by not emptying glass containers at night and by steadily transitioning to **electric trucks** to reduce engine noise.

Performance and impact: The Dutch model has generally been successful. It keeps streets almost devoid of visible trash – you won't see garbage bags piling up on Amsterdam corners. Vermin control is improved (there's no easy food access for rats when waste is underground). Residents also benefit from flexibility; they can drop a small bag whenever convenient instead of storing trash. As one commentary noted, it's nicer to *"carry one small bag out whenever you leave the house, instead of letting a large bag fester for a week then hauling it to the curb"* [35]. However, the system is not without issues: **overflow can occur** if people leave items that don't fit into the bin opening (like a big cardboard box) – these often end up sitting next to the container until a special crew collects them. The city provides separate channels for bulky items or extra waste, but compliance varies. Also, pay-per-drop systems have to ensure they don't encourage *more* illegal dumping (if someone doesn't want to pay, they might dump a bag in a public litter bin or in the woods). Generally, Dutch municipalities have found the **benefits outweigh the downsides**, and the model has spread across nearly all Dutch cities [55] [42].

One particular insight from the Netherlands is the importance of **having sufficient recycling options alongside residual waste bins**. In many Dutch towns, households still use curbside or building-level bins for recyclables (like organics, plastics, etc.) [55]. The underground containers in public are often mainly for **rest waste and glass** – the idea being to minimize unsorted trash. Where fully communal for all fractions, the cities have ensured multiple container types (for PMD – plastic/metal, for paper, etc.) are equally accessible. Pay-by-drop for trash nudges people to use the free recycling bins more. It's a combination of **technology and policy** driving high diversion rates.

For D.C., Amsterdam's example suggests that an **underground container approach can work well in dense urban areas**, yielding very clean streets and potential operational savings. But it requires

investments in specialized trucks and containers, and a robust maintenance regime (those crane systems must work reliably). It also shows that coupling containerization with **user charging schemes** can amplify waste reduction – though that might require legislative changes and strong public communication. Lastly, the Dutch experience underlines tackling side-effects like scavenging; if D.C. has deposit-redemption containers or a population that might rummage in bins, designing containers to prevent that (or providing alternative recycling collection for deposit items) would be wise.

Vienna (and Central Europe): Emphasis on Building-Integrated Collection

Vienna, Austria offers a slightly different take on "shared" waste collection – one that keeps the containers largely **off the streets**. Vienna is renowned for its cleanliness and high waste management standards, yet you won't see many public garbage containers lining sidewalks in residential areas. This is because Vienna's system mandates **waste storage on private property** (usually within each building), combined with a strong city collection service that empties those shared building bins.

Design and setup: In Vienna, virtually every apartment building or housing complex has its own **"Müllraum" (waste room)** or designated bin area, where large standardized containers for different waste fractions are kept [56] [57]. These containers are communal for everyone in that building, but they are not out on the curb except during collection. By law, new buildings must include space for waste separation infrastructure as part of the permitting process [58]. This policy means the city's public spaces stay free of permanent trash bins (improving aesthetics and footpath clearance), unlike Barcelona or Amsterdam where bins are on the street. For smaller buildings or areas where providing full recycling bins on-site isn't feasible, Vienna supplements with **public recycling points** – there are about **4,400 neighborhood drop-off points** for things like glass, paper, and metal packaging across the city [59]. Residents of small buildings might use these for recyclables if they don't have their own bin, but *residual garbage is always collected from the property directly* [60]. Essentially, **all households are enrolled in municipal collection**, and private hauling is rare (only purely commercial premises can opt out) [61].

Operations: The city (Municipal Department 48, MA48) operates a fleet of about **300 collection vehicles with nearly 1,000 crew members** [62], serving the entire metro population (nearly 2 million). The collection is frequent: by law at least once a week for refuse, and often more in big buildings [63]. Crews enter properties or courtyards, roll the containers out to the truck, empty them (either by rear-loading tipper or front-loading for very large dumpsters), and return them to their spot. It's labor-intensive but very thorough. On average **100,000 containers are emptied per day** in Vienna [64] – an impressive figure that shows how much container capacity is deployed citywide (total ~460,000 containers in service for various waste types [65]). Despite this scale, the system runs efficiently through meticulous route planning and that ingrained requirement that buildings make it easy for crews to access the bins.

Economic and policy aspects: Vienna's approach effectively internalizes costs by property. There is a **waste fee based on container volume and pickup frequency** for residual waste [66]. Crucially, **recycling and organic waste collection are provided free of charge** to property owners [67]. This acts as a financial incentive: if a building's residents recycle a lot, they can potentially downsize their residual waste container (or reduce pickup frequency) and save on fees. It's a PAYT principle applied collectively at the building level. Since the city only supplies recycling bins directly to properties that generate enough volume (to ensure they are well-utilized) [68], smaller residences might not have an on-site paper bin, for example – those folks use the public recycling points. But any building with a decent number of units will get the full suite of bins (paper, plastic/metal, organics, etc.) provided for free. This strategy ensures **high participation** in recycling

and keeps per-capita disposal low, because no one wants to pay for an oversize trash bin when they could divert material to free bins.

The economic efficiency of Vienna's method is noteworthy. By having residents consolidate waste in a few shared bins per building, the city avoids having to handle dozens of individual bins per block. And since bins are on private land until collection, there's no need for permanent street infrastructure (besides those 4,400 drop-off sites for extra recyclables). It does rely on a cooperative populace and building managers – e.g. residents must bring their waste to the building's garbage room and sort it properly. Culturally, this has been in place for decades and is well-accepted in Austria.

Outcomes and lessons: Vienna achieves a high recycling rate (around 50% for municipal waste) and sends very little to landfill, thanks to both recycling and an extensive waste-to-energy system for the rest. The streets are famously clean; you rarely see litter around trash containers because the containers aren't out except at collection time, and the city also provides many public litter bins that are emptied regularly for on-the-go trash. One key takeaway is the importance of **planning and regulations**: Vienna's requirement that waste disposal be handled on-premises where generated [58] means new developments cannot ignore the logistics of trash – they must build the "invisible" infrastructure (storage space, access for trucks) from the start. This is a valuable idea for cities like D.C.: updating building codes to ensure adequate waste rooms or staging areas can make a huge difference in how a collection system functions, especially if moving to communal bins.

Another lesson is that **financial incentives align with the system**. Because residual waste costs money per volume in Vienna and recycling is free, both landlords and tenants have motivation to separate waste. It's similar to San Francisco's model (coming next) but executed at a multi-family building scale. This could help address equity concerns – rather than charging individual low-income households for every bag, Vienna's system spreads the cost over the whole building and still rewards reduction behavior overall.

For Washington, adopting Vienna's style would imply focusing on **alley or building-based communal dumpsters** rather than placing containers on every street corner. D.C. already has alleys in many neighborhoods that serve a similar purpose (many D.C. rowhouses share common dumpsters or at least cluster their bins in alleys). Formalizing and optimizing that – possibly requiring landlords to share dumpsters or the city to provide larger containers at alley collection points – could be a way to reap the benefits of containerization without cluttering the sidewalks. It also means investing in **good collection crews and equipment** to handle moving containers around in sometimes tight spaces, a discipline where Vienna clearly excels given their daily volume.

Seoul (and South Korea): Tech-Enabled Communal Collection with Pay-as-You-Throw

Seoul provides a fascinating example of communal waste collection with a strong emphasis on **technology and waste reduction**, particularly for food waste. South Korea in general has one of the most rigorous waste management regimes, driven by necessity (limited landfill space) and law. While general trash in Seoul is disposed of in *prepaid bags* (a form of PAYT where residents buy official bags for any garbage, which they then leave at designated collection points), the city's approach to organics stands out: **RFID-based communal food waste bins** have been rolled out extensively.

System description: In most apartment complexes (where the majority of Seoul's population lives), there are **shared food waste bins** stationed, often near building entrances or in garage areas. These bins are like

high-tech garbage cans: a resident taps an RFID card or key fob, which **weighs the food waste they dump and charges them by the kilogram** [69] [70]. The lid then closes automatically. The fee is modest (in Seoul, about **130 won per kg**, roughly $0.10 per kg of food waste) and is billed to residents, usually as part of monthly maintenance dues [71]. This system was introduced in the 2010s to overcome issues with the previous methods (which involved either buying special food waste bags or using communal bins without measurement). By making disposal data **precise and transparent down to each household** [72] [36], Seoul achieved a remarkable behavioral shift: people became very conscious of how much they were throwing away. Nationally, South Korea now recycles or composts **over 96% of its food waste** [73], and Seoul saw its food waste quantities drop by ~24% over a decade despite population growth [37].

Operations: The food waste bins in Seoul are typically serviced by municipal or contracted crews who collect the accumulated food slurry or scraps from these units. Often, the bins have a built-in compactor or dewatering mechanism to reduce volume and odor. Collection might be every day or several times a week, since food waste is wet and heavy. In some cases, trucks with vacuum hoses suck out the liquid from the bins. Seoul has built large **processing plants specifically for food waste** – one example is the Gangdong district facility, which is built underground and handles 300 tons of food waste per day, fermenting and turning it into biogas, feed, or compost [74] [75]. By separating food waste so thoroughly, they keep it out of landfills and even generate resources.

General waste in Seoul (the non-recyclable remains) must be put in government-sold bags and left out, usually at night, for curb pickup – often just small community piles or curb spots, rather than individual bins. Recyclables are handled through a mix of curbside and drop-off: residents sort paper, bottles, plastics, etc., and typically leave them out on designated days or place them in communal collection areas in their building. It's somewhat labor-intensive (many apartments have a caretaker who tidies up and sorts the recycling area), but it works due to social norms and regulations.

Social and equity aspects: The RFID bin system has been well-received by many, as it directly engages individuals in waste reduction. People in Seoul have "learned tricks" to minimize the weight they toss – e.g. **draining and squeezing moisture out of food scraps** – because they see the weight-based fee every time [36]. It essentially gamified waste reduction. And because one can take out small quantities at any time (the bins are accessible 24/7), it's convenient; no one has smelly food sitting in their kitchen. There's even evidence that families have improved their meal planning and portion control to avoid leftovers once they started seeing the cost of waste [76].

However, there are equity considerations. Initially, the national government funded a lot of these RFID units, but when funding stopped, poorer districts found it hard to buy new ones or expand coverage [48]. As of 2025, about **81.6% of Seoul's apartment residents** were served by RFID food bins, but only ~38% of all housing types (since detached homes or small buildings in older areas still use the bag system) [77]. The city is working to increase coverage, but it shows that high-tech solutions can widen a gap between well-resourced and less-resourced communities if not carefully supported. Also, the elderly or less tech-savvy had to adapt to using a card and digital system – community education and simple interfaces (like voice prompts on the machine) helped bring everyone on board.

Comparison to shared container concepts: Seoul's approach is a mix of communal and individualized strategy. The food waste bins are shared in that an entire building or block uses the same bin, but each person's use is individually measured and billed. This hybrid model could be very relevant to D.C.: imagine

communal dumpsters that log each household's contributions (via a trash cart RFID or a keycode) and charge accordingly – it could drive waste reduction while keeping the convenience of shared infrastructure.

One more point from South Korea is how **policy drives infrastructure**. They banned landfilling of food waste in 2005 and ocean dumping of food waste liquids in 2013 [78], forcing cities to innovate collection and processing. It shows that setting firm regulations (e.g. "no organics in landfill") can catalyze the adoption of new systems like the RFID bins.

For D.C., Seoul's success with RFID food waste collection highlights the potential of technology to achieve environmental goals (nearly 97% recycling of food waste!) [73]. It also underscores that **citizens will respond to price signals and feedback** – when they see in real time how much waste they produce and have a direct incentive to reduce it, many do change habits. The question will be how to translate that to an American context and for other waste streams. Perhaps introducing smart compost bins in neighborhoods, or at least starting with voluntary incentive programs (like giving discounts for those who compost actively), could be pathways. The South Korean model requires an upfront investment in equipment and a cultural willingness to engage at that level of granularity, but its results in waste reduction and resource recovery are undeniable.

San Francisco: A High-Performance Three-Bin Curbside System (Contrast Case)

While not a communal container system, San Francisco often enters the conversation as a leader in municipal waste management. It provides a useful **contrast** to the shared-bin approach: San Francisco has achieved landfill diversion rates around **80%** – one of the highest in the world – using an individualized **curbside "three-bin" system** (black for trash, blue for recycling, green for compost) [79] [80]. Each household or building has its own set of bins collected by the city's contractor (Recology) on a weekly schedule. The question is, could D.C. (or others) reach SF's success without communal containers? And what trade-offs are involved?

System highlights: San Francisco's "Fantastic Three" bins were rolled out citywide by 2000 [80]. The city made recycling and composting **mandatory in 2009** – residents and businesses *must* separate their waste, and trash inspectors (dubbed jokingly "garbage police") can issue fines for non-compliance [49]. To drive participation, SF employs a PAYT approach: **trash bin service costs money, but recycling and compost bins are free of charge** [81]. This means if you reduce your trash enough to switch to a smaller garbage bin, your collection bill goes down – a strong incentive to divert material into the free bins. Many single-family homes indeed chose mini 20-gallon trash cans to save on fees, maximizing what they put in the recycle or compost bins. For businesses, the city even gives **rate discounts for higher recycling/compost diversion percentages** [82], aligning economic benefits with waste reduction.

Operations: SF's model is traditional curbside collection. It requires a fleet of trucks (typically one truck for landfill + compost, and another for recycling, since recycling is sorted at a facility) and lots of labor to roll bins to trucks, etc. It's labor-intensive but very thorough. Recology, the private hauler, invests heavily in outreach and sorting technology to keep improving diversion. The city's approach has also involved extensive **public education** – teaching people how to sort properly, which they've done through school programs, campaigns, and even what they call *"Fantastic 3"* cartoon branding.

Outcomes and implications: By 2019, San Francisco reportedly kept **80% of its waste out of landfill** [79] (this includes recycling, composting, and also things like construction debris diversion). This far exceeds the

U.S. average recycling rate (~35%). Notably, about **590 tons of organic material** (food scraps, yard waste) are collected daily and composted [83] – demonstrating the power of offering citywide compost service. The compost is put to beneficial reuse (e.g. sold to vineyards and farms) [84], creating a circular economy aspect.

The San Francisco example shows that **high diversion is possible without communal drop-off**, but it relied on a combination of factors: strong policy (mandates and bans on certain disposables), economic incentives, a single dedicated hauler with modern facilities, and a culture of environmental consciousness among residents. It also doesn't solve some issues communal systems address – for instance, SF still has to contend with thousands of rolling bins crowding curb-sides on collection days (aesthetically not great, and physically inconvenient on narrow sidewalks). The city has spent effort on designing better public litter cans too, since even with good household service, street litter was a problem (famously, SF piloted **$20,000 designer public trash cans** to reduce litter, showing that keeping streets clean is an ongoing battle even for them).

Why compare SF here? For Washington, D.C., it's instructive to see that a **curbside bin model can achieve excellent environmental results**, but it takes rigorous implementation and perhaps more ongoing effort by residents (separating into multiple bins, following schedules, etc.). A communal system might be less onerous day-to-day for residents (no need to manage multiple bins per house), but the trade-off is the burden shifts to the city to ensure those shared bins are used correctly and maintained. In a way, SF's model places a lot of responsibility on each household (each has to sort their trio of bins properly), whereas a shared system centralizes the responsibility (fewer bins to monitor, and the city can control access or sorting at those bins). San Francisco also had the advantage of a long-standing culture of waste awareness and the leverage of state laws (California AB 939 set diversion goals decades ago [85]).

For D.C., which currently has a system more like SF (curbside bins for houses, dumpsters for large buildings), the decision might come down to urban form: some neighborhoods might benefit from a shift to shared containers if curbside is inefficient there, but others might not. SF's case suggests that **either approach can work** environmentally if paired with strong incentives and education. However, if D.C.'s goals include reducing visual clutter or rodent issues from individual bins, then communal containerization (as in Barcelona/Amsterdam) offers solutions that SF's method doesn't inherently address.

In summary, the comparison cities illustrate a spectrum: from Amsterdam's high-tech communal bins and Vienna's building-centric model, to Seoul's smart pay-by-weight communal approach, and San Francisco's intensive curbside program. All have elements D.C. can learn from:

- Communal systems (Amsterdam, Seoul, Barcelona) show **improved street cleanliness and potential for PAYT incentives**, but need investment in infrastructure and monitoring.
- Building-focused systems (Vienna) demonstrate how **good planning and free recycling services** can integrate waste management seamlessly into urban life.
- Traditional bin systems (San Francisco) prove that with the right policies (mandatory recycling, fee incentives) you can reach very high recycling rates, though they may not solve issues like curb congestion or labor costs.

The best approach for D.C. might well be a hybrid – adopting communal containers in dense commercial or multi-family areas (perhaps underground or in alleys), while maintaining individualized service in suburban-style neighborhoods, all under a unifying framework of waste reduction incentives.

Insights for Washington, D.C.: Scalability, Feasibility, and Preconditions

Considering a rollout of shared containerized collection in Washington, D.C., we distill the following key insights from the above cases:

1. Scalability Requires Commitment: To gain economic and operational efficiency, a communal container system should be implemented at scale in a given area, rather than piecemeal. Barcelona's analysis showed costs only level out when the system is widespread and not run in parallel with old methods [26]. For D.C., this means choosing pilot neighborhoods carefully and planning for full coverage in those zones (or citywide adoption in the long run) to avoid the expense of duplicative routes. Start with a dense area where individual curbside bins are problematic (e.g. downtown, or rowhouse districts with narrow streets) to pilot the approach, but ensure the pilot is large enough to test route efficiency and not just a one-block demo.

2. Infrastructure and Policy Prerequisites: Washington would need to invest in appropriate infrastructure: **containers** (potentially smart ones with RFID if PAYT or tracking is desired), **vehicles** (side-loading or crane trucks for lifting containers), and possibly modified **streetscapes** (dedicated container pads or enclosures, particularly if considering underground units or alley container stations). Additionally, updating **regulations/building codes** would help – similar to Vienna's requirement for on-site waste storage [58], D.C. could mandate new developments include space for communal container pickup or install underground container systems in large projects. If pay-as-you-throw is a goal, then a legal framework for variable-rate trash billing must be in place and technology (smart cards, etc.) distributed to residents.

3. Community Engagement and Site Planning: Street-level feasibility in D.C. will hinge on thoughtful siting of containers and community buy-in. Planners should identify **locations for containers** that balance accessibility (close to residents) with minimal obstruction (not blocking sidewalks or sightlines). Barcelona's move to lower-profile bins [3] and Amsterdam's use of underground bins both address aesthetic and space concerns. D.C. might consider alley locations or repurposed parking spaces for containers – this could spark concerns about parking loss, so engaging the community on trade-offs (cleaner streets and fewer rats vs. a couple of parking spots) is important. **Historic districts** might oppose visible bins, so alternatives like underground containers or small-group "micro stations" could be tested there. Early and transparent communication about the goals (cleanliness, cost savings, equity) and design choices will be needed to avoid the backlash Barcelona initially faced by imposing changes top-down [19].

4. Incentives and Enforcement Mechanisms: A major factor in success is getting residents to use the system properly. D.C. should plan for a combination of **incentives** (e.g. discounts or rebates for households that actively participate in composting, similar to Barcelona's organics fee rebate [11] or San Francisco's bin downsizing savings [81]) and **enforcement** (fines for illegal dumping, surveillance of chronic dump sites, "tagging" of contaminated recycling with warnings, etc.). Implementing a **PAYT program** citywide could dramatically reduce waste, but it might be politically challenging. An alternative first step is what SF and Vienna do: charge based on bin size/collection frequency rather than weight, which can be done without individual tracking. With communal bins, this could mean charging multi-family buildings or neighborhoods by the volume of waste collected (creating peer pressure to reduce waste). If D.C. opts for RFID/bin-access cards, it can more precisely charge per use – but then it needs the tech infrastructure and robust customer account management to bill people, which is a significant administrative project.

5. Maintenance and Rapid Response: Cities with communal containers learned that **maintenance and cleaning are as important as collection**. D.C. will need dedicated crews or contracts to routinely wash the containers (to prevent odor and pest issues) and to repair broken lids, locks, or sensors quickly. A rapid response team to pick up any overflow or illegally dumped items next to containers is also critical – otherwise one or two bad actors can make a whole block look trashy. Essentially, the city has to be ready to take on more direct responsibility for the public waste space, whereas in the individual-bin model residents handle their own bins. This is a manageable shift but requires planning and resources (Barcelona, for instance, allocated more workers to street-cleaning around containers in its new contract [41]).

6. Context-Sensitive Approach: Not all neighborhoods are alike, so D.C. might implement a **hybrid system**. The comparisons suggest communal containers work best in **high-density, multi-family, or mixed-use areas** where individual bins are impractical or unsightly. In suburban-style single-family blocks with yards and alleys, the current individual bin system might remain more practical (or communal bins could be placed in alleys if neighbors agree to consolidate service). Barcelona's approach of using different models in different urban zones [20] is a good precedent. For D.C., this could mean Supercan bins continue in outer neighborhoods, while inner-city wards get shared dumpsters or underground bins. Scalability can also mean scaling by area type – prove it in one context, then adapt to another.

7. Monitor Impacts on Equity: As the system rolls out, D.C. should monitor who is benefiting and who might be adversely affected. For example, if walking to a container is problematic for some elderly residents, perhaps provide an assisted service or neighbor volunteer program. If alley containers attract misuse by non-residents (e.g. contractors dumping debris), enforce strictly to protect residents. Ensure that any cost savings from efficiency are funneled back into either improved service or lower fees for ratepayers, so the public sees a tangible benefit. And if a PAYT model is introduced, be cautious to not unduly burden low-income households – perhaps include a rebate or credit for those on assistance, or start with a revenue-neutral structure that rewards recyclers rather than simply adds fees.

8. Don't Forget the Waste Reduction Goal: Ultimately, the reason to consider these systems is not just operational tidiness, but to reduce waste and increase recycling/composting. In all the cities studied, the containerization was accompanied by broader waste strategies (public education, bans on certain disposables, promotion of reuse, etc.). D.C. should similarly frame communal collection as one piece of a larger zero-waste or sustainability plan. For example, deploying organics containers citywide would pair naturally with D.C.'s composting programs and could significantly divert food waste from landfills (as Seoul and SF demonstrated). Likewise, making recycling drop-off convenient and ubiquitous (like Amsterdam's and Vienna's systems) can boost recycling rates. Setting clear metrics – e.g. "reduce rat complaints by X%" or "increase recycling by Y% in pilot area" – will help measure success and adjust the program.

In conclusion, a shared containerized waste system *could* be a viable and beneficial approach for Washington, D.C., if implemented with careful attention to design, community needs, and complementary policies. The experiences of Barcelona and peer cities suggest that such a system can lead to **cleaner streets, more efficient collections, and potentially better recycling outcomes**, but only if the city invests in the necessary infrastructure, builds public trust and engagement, and remains agile in addressing challenges like misuse and maintenance. Washington, D.C. would need to ensure the system is tailored to its urban form – perhaps starting in the densest neighborhoods – and that it maintains the flexibility to adjust (using data and feedback) as residents acclimate to the new way of handling trash. With thoughtful rollout and strong political/community support, D.C. can leverage these global lessons to modernize its waste collection in a way that is scalable, street-friendly, and aligned with long-term sustainability goals.

Sources:

- Barcelona's color-coded communal bins and recycling system [2] [1]; optimal usage times [14].
- Barcelona's pilot of smart locked organics containers (Sant Andreu), costs and results [10] [11] [31].
- City contract upgrades in 2022 (new bin designs, more electric vehicles, expanded recycling infrastructure) [3] [13].
- ACR+ study of six cities with underground containers: operational, economic, and cleanliness impacts [86] [26]; specific data from Florence showing 36% lower operating cost than door-to-door [22].
- Cleanliness: communal bins vs. bagged trash (less sidewalk litter, though some illegal dumping) [30].
- Social and spatial concerns in Barcelona: resident pushback to door-to-door, need for different solutions by urban typology [19] [38].
- Amsterdam's underground container system and use of RFID access cards [87] [52]; convenience and pest reduction benefits [88].
- Comments on Netherlands system: pay-per-drop using RFID, occasional bags left when malfunctions occur [42].
- NL Times report on Dutch bins being vandalized for deposit containers, causing damage and prompting locks & other measures [43] [44].
- Vienna's building-centric system: number of containers, legal requirement for on-property waste storage, fee structure (residual paid, recyclables free) [60] [67] [58].
- Vienna's collection scale and staffing [64] [62].
- Seoul's RFID food waste program: near-universal apartment coverage, weight-based fee, huge recycling rate (~97%) and waste reductions achieved [73] [89].
- Behavioral change from real-time weight tracking in Seoul [36] and the necessity of continued funding for expansion [48].
- San Francisco's 80% diversion success with mandatory recycling/composting and financial incentives (free recycle/compost bins, pay for trash size) [79] [81].
- San Francisco's historical policy drivers (state law AB 939) and "garbage police" enforcement of sorting [85] [49].

---

[1] [2] [14] How recycling works in the streets of Barcelona and Spain
https://www.skipthelinebarcelona.com/recycling-in-barcelona/

[3] [12] [13] [15] [21] [24] [29] [34] [41] Nuevos contenedores, menos ruido... así es la recogida de residuos que Barcelona empieza a implantar este lunes
https://www.20minutos.es/noticia/4966626/0/nuevos-contenedores-menos-ruido-asi-es-recogida-residuos-barcelona-empieza-implantar-este-lunes/

[4] Sistemas/modelos de recogida selectiva. Agència de Residus de Catalunya
https://residus.gencat.cat/es/ambits_dactuacio/recollida_selectiva/residus_municipals/materia_organica_form_-_fv/recollida_selectiva/els_sistemes_models_de_recollida_selectiva/

[5] Cómo funciona la recogida neumática de residuos - Tomorrow.City
https://www.tomorrow.city/es/mas-limpia-mas-silenciosa-mas-barata-como-funciona-la-recogida-neumatica-de-residuos/

[6] Barcelona's Envac system is being modernised
https://www.envacgroup.com/news/barcelonas-envac-system-is-being-modernised/

7 8 9 10 11 19 20 31 33 38 39 Barcelona estrena sus primeros contenedores de basura inteligentes en Sant Andreu
https://www.lavanguardia.com/local/barcelona/20220117/7993106/barcelona-estrena-primeros-contenedores-inteligentes-sant-andreu.html

16 Ultrasonic bin sensors monitor waste levels in Mollerussa, Spain
https://www.sensoneo.com/success-stories/monitoring-waste-mollerussa-spain/

17 18 22 23 25 26 27 28 30 32 86 circulareconomy.europa.eu
https://circulareconomy.europa.eu/platform/sites/default/files/2025-04/ACR-beneath-surface-executive-summary.pdf

35 42 50 51 52 53 54 55 87 88 Amsterdam's Smart System of Underground Garbage Bins - Core77
https://www.core77.com/posts/102208/Amsterdams-Smart-System-of-Underground-Garbage-Bins

36 37 48 70 71 72 73 74 75 76 77 78 89 'You learn tricks to reduce it': the smart bins measuring food waste in South Korea | South Korea | The Guardian
https://www.theguardian.com/environment/2025/dec/18/smart-bins-measuring-food-waste-south-korea

40 Los vecinos se niegan a usar los contenedores digitales que ...
https://www.instagram.com/reel/DKj2OMNtfLz/

43 44 45 46 47 Amsterdam has most smashed trash bins of all Dutch cities as people hunt cans, bottles | NL Times
https://nltimes.nl/2025/10/28/amsterdam-smashed-trash-bins-dutch-cities-people-hunt-cans-bottles

49 79 80 81 82 83 84 85 Garbage police who walk the beat
https://impact.economist.com/projects/foodsustainability/blogs/san-francisco/

56 57 58 59 60 61 62 63 64 65 66 67 68 wien.gv.at
https://www.wien.gv.at/pdf/ma48/infoblatt-sammlung-en.pdf

69 [PDF] Pay as you throw system of Seoul
https://seoulsolution.kr/sites/default/files/policy/2%EA%B6%8C_Environment_Pay%20as%20you%20throw%20system%20of%20Seoul.pdf